

Design Document

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TRINITY GAMING

WAVE SURVIVAL · SOLO & SQUAD CO-OP · PC & MOBILE

Preface

Zombie Rush started from a simple question: what would a round-based zombie shooter feel like if it was made for a phone as much as for a PC? We have always enjoyed horde shooters, from the Zombies mode of Call of Duty to Killing Floor 2 and Dead Trigger 2, and we wanted to make one of our own that keeps their tension but is quick to pick up, plays well on a touch screen and is fun with friends. The safe base and the expanding fight zone grew out of that idea: a place to catch your breath between waves, and a battlefield that gets bigger every time the players push their luck.

This document is the design of Zombie Rush as it stands today. A large part of the game is already built in Unity: the player controller, the weapons, the safe base, the waves, the zombies and the expanding fight zone, and every number in the mechanics chapters is taken from the current build. The parts still to come, such as the new zombies, the full arsenal, audio and online squad co-op, are marked as planned or proposed, so that every later piece of work has a clear target.

We hope you enjoy this document.

Trinity Gaming

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Figure 1: Zombie Rush key art: the ruined city at night.



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Game Concept

Introduction

Zombie Rush is a first-person wave-survival shooter for PC and Android. The player is a survivor holding a walled-off safe base at the edge of a ruined city, where every night the dead rise out of the ground in ever-growing waves.

Background

Zombie Rush is a new, original game by Trinity Gaming. It takes the round-based horde mode made popular by the Zombies mode of Call of Duty and rebuilds it around a safe base and an expanding fight zone, with runs short enough to play on a phone. The game is built in Unity 6 with the Universal Render Pipeline, so a single project can target both Windows and Android.

Zombie Rush is made by a small team, so the art pipeline is kept small as well. The zombies start as concept art and are turned into 3D models with Meshy and animated with Mixamo, and the ruined city is built from a post-apocalyptic building pack from the Unity Asset Store.

Zombie Rush is still in development. This document describes the game as it is built today, together with the features that are already planned, and more zombie types, weapons and features will be added later as development continues.

Description

Like the horde shooters that inspired it, Zombie Rush drops the player into a fight that only gets harder. Each run starts inside the safe base, where the zombies can never enter and the stations hand out weapons, armour, ammunition and medkits. The moment the player steps out, the wave begins and the zombies rise out of the ground around them. It is the player's job to survive each wave, collect the coins the zombies drop, and decide whether to spend them on pushing the fight zone outward, knowing that more room also means more ground to defend. The run lasts until every player is down, and the only question is how many waves they can survive.

Key Features

Endless Waves:

Zombie Rush has no final wave. Every wave adds more zombies and makes each one tougher and harder-hitting than the wave before, so every run ends in a last stand.



Safe Base and Fight Zone:

A safe base the zombies cannot enter gives the players a place to rearm, heal and regroup. In front of it, the fight zone grows outward as the players spend their coins, opening up more of the city to fight in.

The Horde:

Four zombie types, the Walker, the Yaku, the fast Feral and the heavy Miner, rise out of the ground all around the players, so they are approached from many directions at once. More zombie types will be added later in development.

Arsenal:

The Pistol and the Assault Rifle are carried from the start, the Machine Gun and the Shotgun are special weapons, and every player gets a grenade at the start of every wave. Further guns are earned through progression.

Solo or Squad:

Zombie Rush can be played alone or with up to three friends. The number of zombies, their health and their spawn speed scale with the size of the squad.

Graphics:

Realistic zombie models and a ruined city at night, with a moonlit sky, fog and dark post-processing, give the game a grim, cinematic look on PC and on mobile.

Difficulty:

The difficulty is chosen before each run on three levels, Easy, Normal and Hard, together with the time of day: a moonlit night or daylight.

Genre

Zombie Rush is a first-person shooter in the wave-survival, or "horde mode", genre, with a light economy and progression layer. It sits between the arcade horde shooters of PC and console, where points earned from kills open up the map, and the pick-up-and-play zombie shooters that are popular on mobile. Runs are short, the rules are learned in seconds, and the challenge comes from the rising pressure of each wave.

Platform

The targeted platforms are PC (Windows) and Android, released through Google Play, with both versions built from the same Unity project. PC, played with keyboard and mouse, is our primary platform; the Android version uses touch controls and targets 60 frames per second.



Concept Art



Figure 2: The horde: Walker, Feral and Miner.



Figure 3: The Walker



Figure 4: The Feral



Figure 5: The Miner



Game Mechanics

Core Game Play

The core game play in Zombie Rush is a loop of surviving a wave, collecting its rewards and spending them to make the next wave survivable. Everything the player does serves one of those three steps.

Fighting – Combat is about positioning under pressure. Zombies rise out of the ground 10 to 22 metres from the players and head for the nearest player outside the base. A player who stands still is surrounded within seconds. The ruined buildings break the zombies' lines of approach and create choke points, and the zombies path around obstacles with the NavMesh, so good use of cover buys real time. The pistol rewards accuracy at range, the shotgun deletes anything that gets close but reloads slowly, and the grenade clears a crowd. Choosing when to reload, and when to switch weapons instead, is a constant decision.

Economy – Kills drop coins and, less often, ammunition. Between waves the players decide whether to spend coins on expanding the fight zone. A bigger zone gives more space to retreat into and reaches more cover, but it also means more ground to watch.

The Base – The safe base is a fixed refuge. Zombies cannot enter it and they guard its edge, so players can always retreat to rearm, heal and pick up armour. The price of safety is time: a wave does not end while zombies are still alive outside.

Game Flow

A typical run follows the same rhythm, wave after wave:

- **Deploy** – the players start inside the safe base and can take weapons, armour, ammunition and a medkit from the stations. Nothing spawns until a player steps out of the base.
- **Announcement** – "WAVE N INCOMING" appears for 3 seconds and the players are healed back to full health.
- **Spawning** – zombies rise out of the ground 10 to 22 metres around the players, in rounds 0.8 seconds apart, until the wave's total has spawned.
- **Fighting** – the wave continues until every zombie is dead. Each kill drops coins and sometimes ammunition.
- **Clear** – "WAVE N CLEARED" appears, followed by a 5-second breather to return to the base, rearm and spend coins.



– **Next wave** – the wave number goes up and the loop repeats. When every player is down, the results card is shown.



Figure 6: In game: a Miner charges at the player during wave 1, while the safe base wall glows behind him.

The size and strength of each wave follow two simple formulas: zombies per wave = 5 + 2 × (wave − 1), and zombie strength multiplier = 1 + 0.15 × (wave − 1). The multiplier scales zombie health and damage, and scales movement speed up to a cap of 1.6×. In a squad the zombie count, zombie health and spawn speed are scaled again (see Multiplayer).

Wave	Zombies (Solo)	Zombies (Squad of 4)	Strength multiplier	Walker HP / hit damage	Yaku HP / hit damage	Walker / Yaku run speed (m/s)
1	5	14	×1.00	60 / 12	–	3.5 / –
2	7	20	×1.15	69 / 13.8	126.5 / 20.7	4.0 / 3.7
3	9	26	×1.30	78 / 15.6	143 / 23.4	4.5 / 4.2
5	13	37	×1.60	96 / 19.2	176 / 28.8	5.6 / 5.1
10	23	65	×2.35	141 / 28.2	258.5 / 42.3	5.6 / 5.1

From wave 5 onward the Walker runs at 5.6 m/s, faster than the player's 5 m/s walk, so from that point the players must sprint to open distance. This is the intended difficulty curve: early waves teach the loop, later waves force movement.



Characters

The Survivor

The player character is an unnamed survivor seen only in first person. The survivor has no special abilities: 100 health points, up to 100 points of armour that soak up 66% of incoming damage while they last, and a loadout of a pistol and an assault rifle. Health is restored to full at the start of every wave, so each wave is a fresh test rather than a slow drain; armour and medkits are collected from the stations at the base. The survivor walks at 5 m/s, sprints at 8 m/s, crouches at 2.5 m/s and can jump about 1.2 metres, enough to clear low props.

Zombies

Four zombie types make up the horde. All of them share the same AI, health and collision setup; each type only has its own model, animations and stats asset. The Walker, the Yaku and the Miner stats below are the values in the current build. The Feral is a new design: its stats are proposals for balancing and its 3D model will be provided later. Every kill table shows how many hits it takes to kill that zombie at different waves; "–" means that zombie type does not spawn in that wave.

Zombie	Role	First wave	Data
Walker	The baseline zombie: common, steady, scales with the wave	1	Implemented
Yaku	The tank: more health and a harder punch, sprints when close	2	Implemented
Feral	The runner: fragile and very fast, punishes standing still	4	Proposed
Miner	The brute and mini-boss: slow, huge health, crushing hammer	5	Implemented

Planned zombie types (proposals):

- **Spitter** – keeps its distance and throws acid, forcing the players out of cover.
- **Bloater** – explodes when killed, damaging everything near it, so it has to be shot at range.



The Walker



Figure 7: The Walker.

The most common zombie: a pale, bald corpse in a torn grey shirt and trousers, one of the first people to turn. He is slow to start but there are always many of him, he heads straight for whoever is outside the base, and he gets stronger every wave. His punch deals 12 damage every 1.1 seconds at wave 1.

Stat	Walker
Health	60
Walk / run speed (m/s)	2.2 / 3.5
Attack damage	12
Attack range	1.6 m
Attack cooldown	1.1 s
Sight range	25 m
First wave / spawn weight	1 / 3
Data status	Implemented

Hits needed to kill (pistol assumes every shot hits; shotgun runs from all 8 pellets hitting at close range to 5 of 8 at medium range):

Weapon	Wave 1	Wave 3	Wave 5	Wave 10
Pistol	3	4	4	6
Assault Rifle (proposed)	3	4	5	8
Shotgun	1-2	2	2-3	2-4
Zombie health	60	78	96	141



The Yaku



Figure 8: The Yaku.

A heavily tattooed gangster with a sword still stuck through his body. The Yaku first appears in wave 2 and is the tank of the horde: nearly twice the health of a Walker and a harder punch. At distance he walks at 2 m/s; once within 12 metres he breaks into a run at 3.2 m/s, so he punishes players who let him close in. He attacks by crouching, lunging forward to grab with both hands and biting; the damage lands at the moment of the bite, so a quick step back can still save the player.

Stat	Yaku
Health	110
Walk / run speed (m/s)	2 / 3.2
Attack damage	18
Attack range	1.7 m
Attack cooldown	1.4 s
Sight range	40 m
First wave / spawn weight	2 / 2
Data status	Implemented

Hits needed to kill (pistol assumes every shot hits; shotgun runs from all 8 pellets hitting at close range to 5 of 8 at medium range):

Weapon	Wave 1	Wave 3	Wave 5	Wave 10
Pistol	–	6	8	11
Assault Rifle (proposed)	–	8	9	13
Shotgun	–	2-4	3-4	4-6
Zombie health	–	143	176	258.5



The Feral



Figure 9: The Feral.

A long-haired, half-naked infected, covered in blood and moving in a hunched, twitching run. The Feral is fragile but very fast. He appears from wave 4 to break up the rhythm of the horde and to punish players who stand still or reload in the open. He dies to a couple of shots, but he is on top of the player in seconds.

Stat	Feral
Health	45
Walk / run speed (m/s)	3 / 5.2
Attack damage	9
Attack range	1.5 m
Attack cooldown	0.8 s
Sight range	35 m
First wave / spawn weight	4 / 2
Data status	Proposed

Hits needed to kill (pistol assumes every shot hits; shotgun runs from all 8 pellets hitting at close range to 5 of 8 at medium range):

Weapon	Wave 1	Wave 3	Wave 5	Wave 10
Pistol	–	–	3	5
Assault Rifle (proposed)	–	–	4	6
Shotgun	–	–	1-2	2-3
Zombie health	–	–	72	105.7



The Miner



Figure 10: The Miner.

A hulking undead miner in a hard hat with a lamp, a red plaid shirt and heavy overalls, dragging a huge sledgehammer. The Miner is the brute of the horde: very high health, slow, and a hammer blow takes a third of a player's health. He appears from wave 5 and is meant to be a mini-boss that the squad has to focus on together. He walks with a heavy, dragging limp and charges at 2.5 m/s once he has a target, his footsteps shaking the ground. His attack is an overhead sledgehammer slam: a slow wind-up gives the player a moment to step back, and the damage lands only when the hammer hits the ground.

Stat	Miner
Health	320
Walk / run speed (m/s)	1.5 / 2.5
Attack damage	30
Attack range	2.8 m
Attack cooldown	2.7 s
Sight range	40 m
First wave / spawn weight	5 / 1
Data status	Implemented

Hits needed to kill (pistol assumes every shot hits; shotgun runs from all 8 pellets hitting at close range to 5 of 8 at medium range):

Weapon	Wave 1	Wave 3	Wave 5	Wave 10
Pistol	–	–	21	31
Assault Rifle (proposed)	–	–	26	38
Shotgun	–	–	8-12	11-17
Zombie health	–	–	512	752



Figure 11: Miner concept art: the sledgehammer pose that defines his attack animation.



Game Play Elements

Safe Base

The safe base is a half-circle zone of 38 metres in front of the base building, with its straight edge behind the building. It is fixed: it never grows. Zombies cannot enter it and they stop at its edge and wait, facing the base. Inside it, players are safe and use the base stations. The first player to leave the base starts the wave.

Base Stations

The base building carries four kinds of station. They are triggered by walking into them and some recharge after use:

- **Weapon racks** – hand out a weapon from the loadout and never run out.
- **Armour** – adds armour points; the armour then soaks up 66% of incoming damage until it is used up. Recharges after 25 seconds.
- **Ammunition** – refills the reserve of the carried weapons.
- **Medkit** – restores health between waves.

Fight Zone and Spawn Pads

The fight zone is the half circle in front of the base where the waves are fought. It starts at a radius of 58 metres and each expansion adds 15 metres, up to 298 metres, for 20 coins each. Zombies surface close to the players: each one rises at a random spot 10 to 22 metres from a player who is outside the safe base, in the open, inside the zone and with a clear path to that player. A spawn pad glows red-orange on that spot, so the players can read where the next zombie will surface. Zombies start below the ground and rise to the surface over 1 second, during which they cannot move or attack. If every player is inside the safe base, the pads are spread along the arc of the zone at 85% of its radius instead.

Coins

Coins are the only currency. Every zombie killed drops 5 to 15 coins in the current build; the release values will be tuned so that an expansion is a meaningful choice. The coin total and the cost of the next expansion are shown on the HUD.

Ammunition Drops

Each kill has a 25% chance to drop 4 to 10 rounds, which are added to the reserve of the weapons the player carries, up to each weapon's maximum reserve.



Health and Armour

The player has 100 health points. Health is restored to full when each new wave is announced, and a medkit at the base restores it in the middle of a wave. Armour holds up to 100 points and absorbs 66% of each hit until it is used up. Taking damage flashes the screen red.

Day and Night

Before a run starts, the players choose the time of day on the mode screen. Night is the default look: a dark, natural night with a moon, stars, fog and a dark post-processing profile. Day uses bright sunlight. The rules of the game are the same in both.

Weapon Properties

The arsenal has three tiers. The **Pistol** and the **Assault Rifle** are the default loadout every player starts with. The **Machine Gun** and the **Shotgun** are the special weapons, which players can buy in the in-game store. The **Grenade** and further guns are earned through progression: every player gets one grenade at the start of every wave, and more guns unlock as the best wave and the star rating grow.

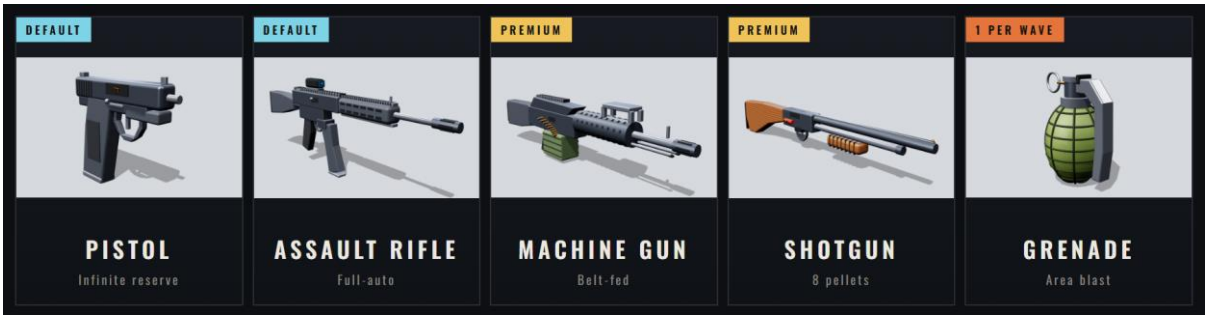


Figure 12: The arsenal: reference renders of the five weapon models that are being built.



Firearms

Pistol and shotgun values are taken from the current build. The assault rifle and machine gun values are proposals for balancing.

Weapon	Pistol	Assault Rifle (proposed)	Machine Gun (proposed)	Shotgun
Tier	Default	Default	Special	Special
Slot key	1	2	3	4
Damage per shot	25	20	16	9 per pellet × 8 (72 max)
Shots per second	6	10	12	1.1
Effective range	100 m	80 m	70 m	18 m
Spread	None (hitscan)	±1.5°	±3°	±6° cone
Magazine	12	30	100	6
Maximum reserve	60	210	300	24
Reload time	1.4 s	2.0 s	4.5 s	2.2 s
Move while shooting	Yes	Yes	Yes	Yes
Auto-reload on empty	Yes	Yes	Yes	Yes

Grenade and Progression Guns

Item	How the player gets it	Role	Proposed stats
Grenade	1 at the start of every wave	Clears a crowd or a Miner	150 damage at the centre, 6 m blast, 2.5 s fuse
Revolver	Unlocked by best wave	Heavy single shots	70 damage, 1.8 shots/s, 6 rounds
SMG	Unlocked by best wave	Fast close-range fire	12 damage, 15 shots/s, 35 rounds
Sniper Rifle	Unlocked by best wave	Long-range precision	150 damage, 0.7 shots/s, 5 rounds

The pistol is the reliable all-rounder: accurate at any range, quick to fire and quick to reload, but it needs several hits per zombie. The assault rifle is the main weapon for general fighting. The shotgun fires eight pellets in a spread and can kill an early-wave zombie in one shot at close range, but it has only six shells in the magazine and a slow reload. The machine gun trades a long reload for a huge magazine and sustained fire. The special weapons are sidegrades with their own handling, and the progression guns can only be earned in play.



Game Physics and Statistics

Movement – The survivor uses a character controller. Walking speed is 5 m/s. Holding Left Shift while moving forward sprints at 8 m/s; sprinting is not possible while crouched or moving backwards. Crouching lowers the body from 1.8 to 1.0 metres and the camera from 1.6 to 0.9 metres, and slows movement to 2.5 m/s. A jump reaches about 1.2 metres under a gravity of 20 m/s^2. Look is clamped to 85 degrees up and down. The edge of the fight zone is applied after movement every frame, so the player slides along the edge rather than passing through it.

Combat – The firearms are hitscan: each shot or pellet is a ray cast from the centre of the camera, so what is under the crosshair is what gets hit. Zombies currently take the same damage wherever they are hit; head-shot bonus damage is planned. A zombie that is hit plays a hit-reaction animation, and a zombie that dies plays its death animation, drops its rewards and is returned to the pool after 4 seconds. Zombies are pooled, so no zombie is created or destroyed during play. Zombie strength scales with the wave as described under Game Flow.

Armour – Armour absorbs 66% of every hit while it lasts, so 100 armour points soak up roughly 150 points of incoming damage and leave the rest to health.

Artificial Intelligence

The zombie AI is deliberately lightweight so that dozens of zombies can run on a phone. Every zombie runs the same short set of rules, and path-finding uses the NavMesh baked for the level:

- **Target** – each zombie picks the nearest living player who is outside the safe base. If every player is inside the base, the zombie walks to the base edge and waits there, facing the base.
- **Attack** – within attack range the zombie stops, turns to face the target and punches on a cooldown, dealing its damage multiplied by the wave multiplier.
- **Chase** – otherwise it follows the NavMesh path to the target, walking if it is further than its run distance and running once closer. The path is recalculated every 0.5 seconds.
- **Base edge** – the zombie is never allowed to enter the safe base; the movement code stops it at the edge.
- **Unsticking** – if a zombie has moved less than 0.15 metres in half a second, it walks in a random direction for 1.2 seconds to escape the corner it is stuck in, then repaths.

Path recalculation is spread across frames so that the cost does not spike when a whole round of zombies rises at once.



Player Controls

Action	PC (keyboard & mouse)	Android (touch)
Move	W A S D	Virtual stick, left side
Look	Mouse	Drag on right half of screen
Fire	Left mouse button (hold)	Fire button
Reload	R	Reload button
Switch weapon	1 / 2 (3 / 4 planned)	Tap weapon icon
Throw grenade (planned)	G	Grenade button
Sprint	Left Shift	Push stick fully forward
Crouch	Left Ctrl or C	Crouch button
Jump	Space	Jump button
Expand the fight zone	P	Expand button
Use a base station	Walk into it	Walk into it
Pause	Esc	Pause button

Multiplayer

Zombie Rush has two modes, chosen on the mode screen: **Solo** and **Squad** (up to four players defending the same base and zone). Phase 1, the mode selection and difficulty scaling, is built; phase 2, the networking with Unity Netcode for GameObjects, Relay and Lobby (no dedicated servers), is planned.

Scaling – the number of zombies, their health and the speed of spawning scale with the number of players, so a wave is not trivial with four guns:

Players	Zombies per wave	Zombie health	Time between spawns
1 (Solo)	×1.0	×1.0	×1.00
2	×1.6	×1.2	×0.80
3	×2.2	×1.4	×0.67
4 (Squad)	×2.8	×1.6	×0.57

- **Authority** – one player hosts the match and decides waves, spawns, deaths, coin drops and the zone radius; the others receive the results.
- **Shared base and zone** – one safe base for the whole squad, and the wave starts when the first player leaves it. Each player earns their own coins, but any player can pay to expand the zone. A player at zero health is downed and can be revived; the run ends when every player is down.



Interface

Flowcharts

The following charts show the flow between screens, the core game loop, the zombie AI and the options screen. The core game loop chart shows how the base, the waves and the fight zone fit together.

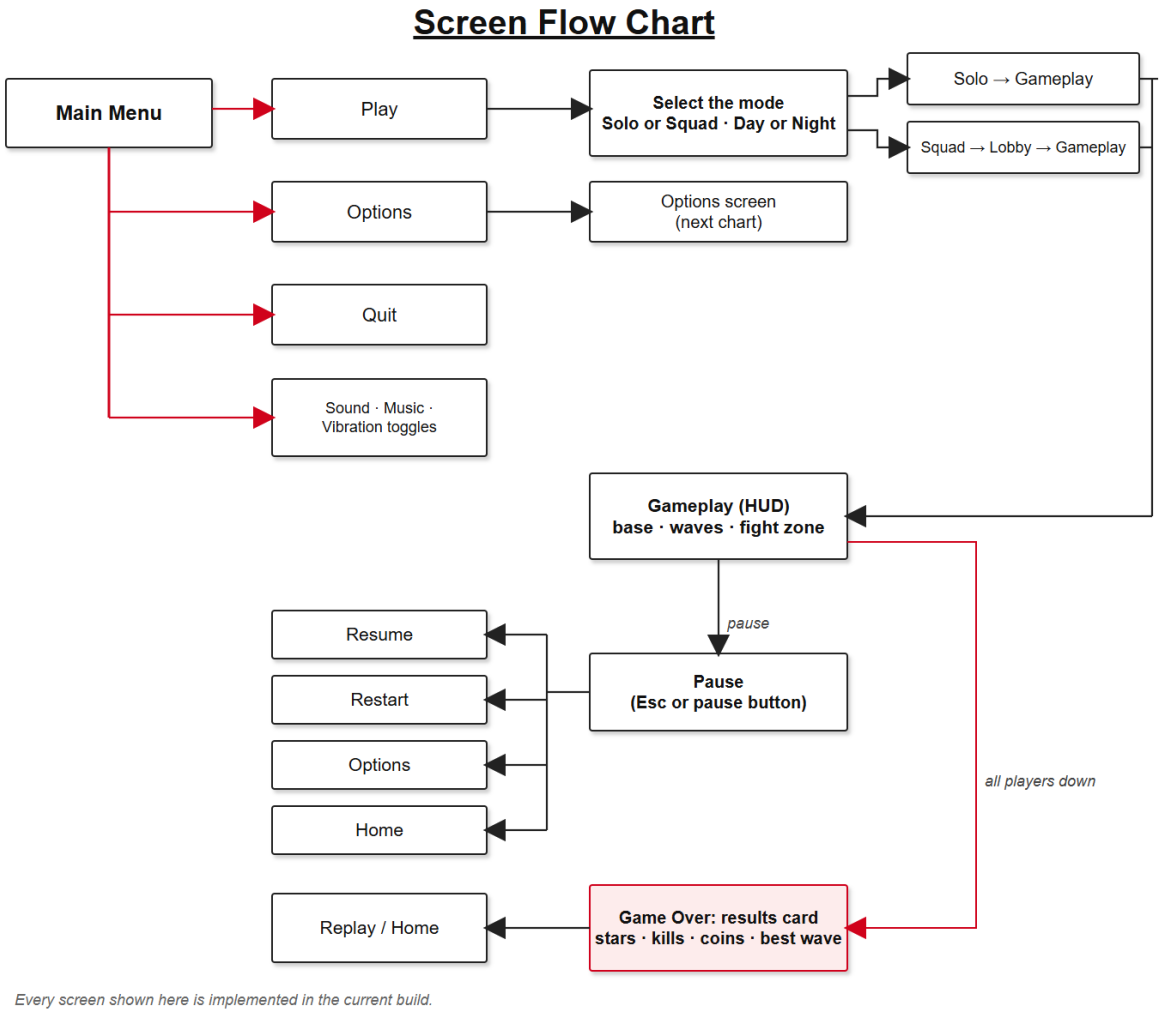


Figure 13: Screen flow.



Core Game Loop Flow Chart

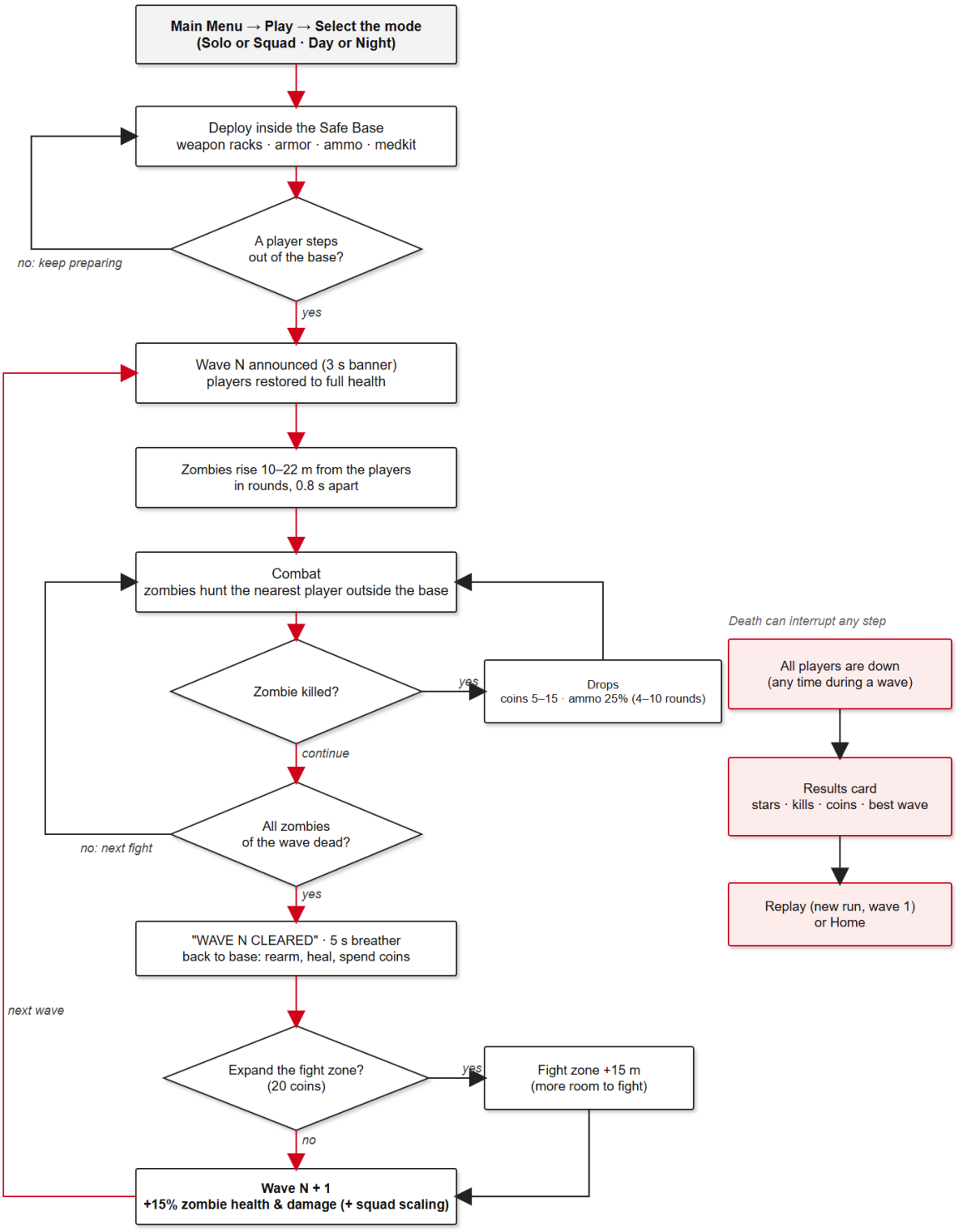


Figure 14: Core game loop.



Zombie AI Flow Chart

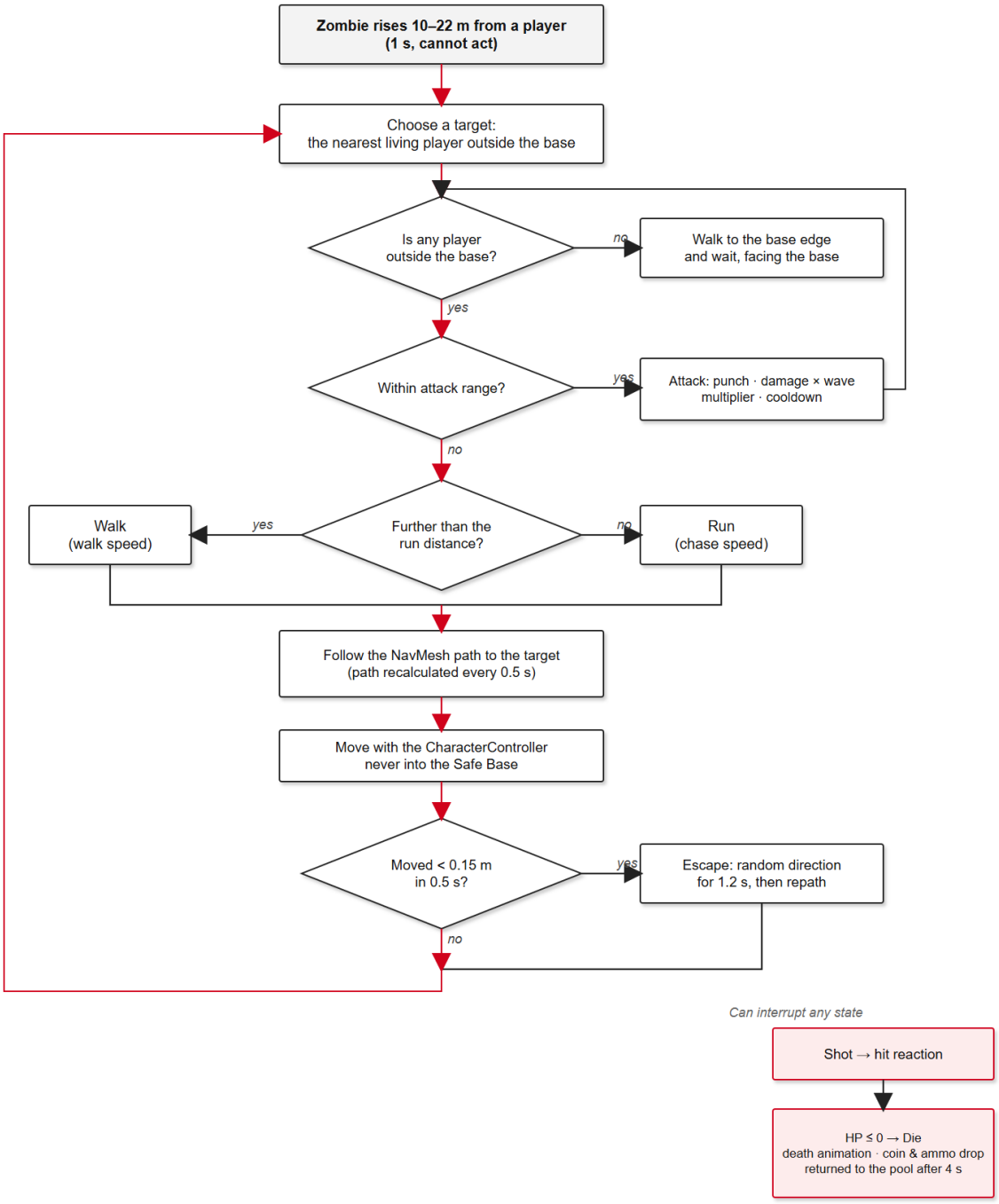


Figure 15: Zombie AI.



Options Screen Flow Chart

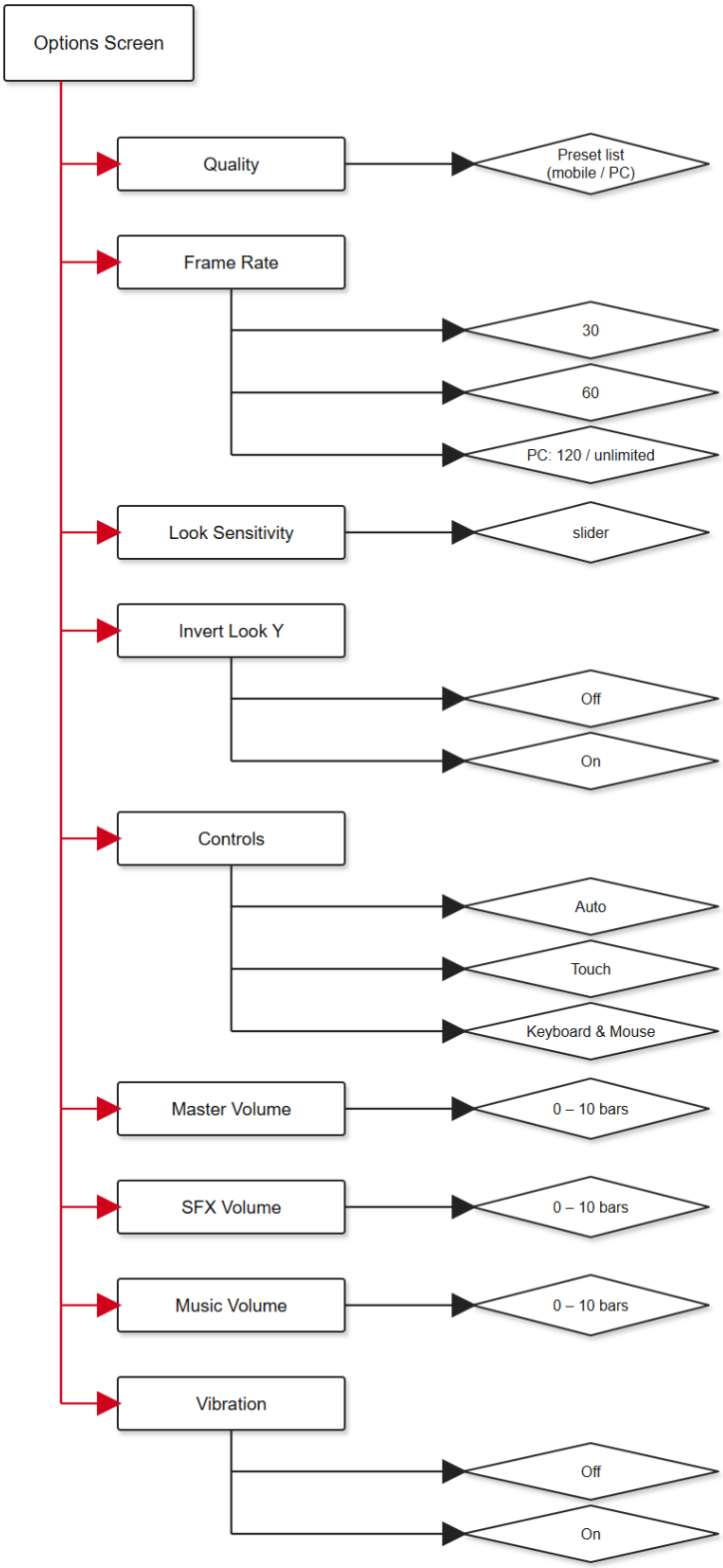


Figure 16: Options screen.



Functional Requirements

Main Menu

The main menu uses a simple highlighted selection that works with the mouse, keyboard or touch.

1:1 – Play

Opens the mode screen.

1:2 – Options

Opens the options screen: graphics quality, frame rate, look sensitivity, invert look, control scheme, master, sound effects and music volume, and vibration.

1:3 – Quit

Closes the game on PC; on Android it returns to the home screen.

Mode Screen

2:1 – Solo or Squad

Two large cards. Solo starts straight away; Squad opens the lobby for up to four players.

2:2 – Day or Night

The time of day of the level. Night is the default.

2:3 – Difficulty

Easy, Normal or Hard.

In Game HUD

3:1 – Zone

Top left, cyan: current radius and maximum, for example "ZONE 73m / 298m", and a bar that fills as the zone grows.

3:2 – Wave

Top centre: "WAVE N" and the number of zombies left, or "INCOMING" and "CLEARED" between waves.

3:3 – Coins

Top right, gold: the coin total. The cost of the next expansion shows in the expand prompt.

3:4 – Kills

Under the coins: the number of zombies killed in this run.

3:5 – Crosshair

A thin cross with a centre dot, always in the middle of the screen.



3:6 – Health and armour

Bottom left: a red health bar that eases down smoothly when hit, with the value over it, and a blue armour bar above it.

3:7 – Weapon and ammo

Bottom right: weapon name, a large magazine count, then the magazine size and reserve. The weapon icons show the other slots. A reload bar fills while reloading.

3:8 – Wave banner

Centre: "WAVE N INCOMING" for the 3-second announcement and "WAVE N CLEARED" afterwards.

3:9 – Damage flash

A full-screen red flash that fades out whenever the player loses health.

Pause and Game Over

Pressing Esc, or the pause button on a phone, freezes the game and opens the pause menu with Resume, Restart, Options and Home. When every player is down, the results card is shown over a dark red screen: a row of five stars (earned at waves 1, 3, 5, 8 and 12), the best wave, the kills and the coins earned, with Replay and Home buttons.



Figure 17: The pause menu.



Figure 18: The results card when every player is down.

Mock Up Screens



Figure 19: Main menu.

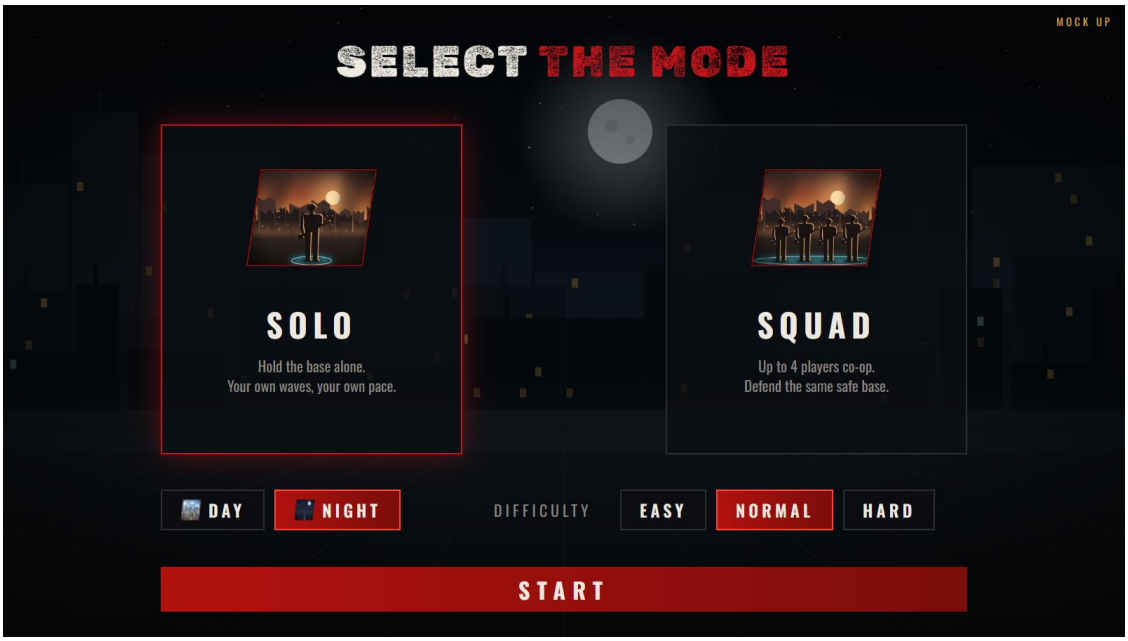


Figure 20: Mode screen: Solo or Squad, Day or Night, difficulty.





Figure 21: Desktop HUD. Numbers match 3:1 to 3:7 above.



Figure 22: Mobile HUD with the touch controls.





Figure 23: Wave announcement.



Figure 24: Pause menu



Figure 25: Results card



Art and Video

Overall Goals

The art direction of Zombie Rush is grounded and gritty: a ruined city at night, with dark buildings, wet streets, fog and a pale moon. The palette is kept dark and desaturated so that the few strong colours the game uses always carry meaning.

That colour language is the most important visual rule. Cyan means safety: the safe base and the zone readout are cyan. Red means danger: spawn pads glow red-orange, blood is dark red, and the screen flashes red when the player is hurt. Gold means reward: coins and the expansion prompt are gold. A player should be able to read the state of the game from colour alone, even on a small phone screen in bright light.

The zombies are realistic rather than cartoonish, with clear silhouettes that read at a distance: the pale bald Walker, the long hair of the Feral, the hard hat and sledgehammer of the Miner, the tattoos and sword of the Yaku. Every visual choice is also checked against the mobile budget, so real-time lighting is kept simple and effects are pooled rather than created on the fly.

2D Art

- HUD elements: health and armour bars, crosshair, weapon and ammo readout, coin counter, wave banners
- Touch control buttons and virtual stick
- Menu screens, mode cards (Solo, Squad, Day, Night) and the Zombie Rush logo
- Weapon and zombie icons; the results card with its stars
- Zombie concept art for the Walker, the Feral and the Miner
- Google Play store icon, feature graphic and screenshots
- Store page art for the PC release

3D Art & Animation

- Environment: post-apocalyptic city buildings, the base building and its stations, a custom night sky and fog
- Zombie models: the Walker, the Yaku and the Miner; the Feral model will be provided later. The Yaku and Miner textures were reworked with grime, decay and blood, and the Miner's helmet lamp glows



- Zombie animations: the Walker uses Mixamo clips (Idle, Walking, Run, Punching, Reaction Hit, Dying); the Yaku and the Miner have their own custom animation sets made in Unity (idle, walk, run, attack, hit) that match each character: the Yaku twitches, staggers and lunges to bite, the Miner limps, charges and slams his hammer
- Foot placement keeps every zombie's feet on the ground, also on kerbs, ramps and rubble
- First-person weapon models: pistol, assault rifle, machine gun, shotgun and grenade, built from the reference renders in this document
- First-person arms with fire and reload animations (planned)

Cinematics & VFX

- Blood bursts on every hit and on death, drawn from a pool of reusable particle systems
- Pulsing glow on the spawn pads and a glowing edge on the safe base and the fight zone
- Zombies rising out of the ground when they spawn
- Camera shake from the Miner's footsteps and hammer slam
- Death animation or collapse when a zombie is killed
- Grenade blast, muzzle flashes and shell ejection
- Full-screen damage flash and the fading wave banners
- A short gameplay trailer for the store pages (planned)



Sound and Music

Overall Goals

The current build has no final audio yet; this chapter sets out what the audio needs to do. Sound in a horde shooter is information as much as atmosphere. The player must be able to hear where zombies are coming from, that a wave is about to start, and that their magazine is about to run dry, without looking at the HUD.

Music should build with the wave: quiet and tense during the breather, rising as the wave is announced, and at full intensity while the zombies are up. On mobile, music is streamed and effects are compressed, so the audio stays small in memory and in download size.

Sound Effects & Music

- Weapons: distinct pistol, rifle, machine gun and shotgun shots, dry-fire click, reload sequences, shell drop, grenade throw and blast
- Zombies: groans and snarls with several variations per type, footsteps, punches, hit reactions, death; a heavy thud for each Miner step
- Spawning: an earth-cracking rumble from each spawn pad as zombies rise, audible in 3D so the player can locate it
- Player: footsteps on pavement and rubble, breathing when low on health, hurt sounds
- Base and economy: station use, coin pickup, zone expansion hum, wave incoming and wave cleared stingers
- Music: layered tracks that add intensity as each wave progresses



Story

Story Overview

Zombie Rush is an arcade survival game, so story is kept to a premise that explains the setting. The premise below is a proposal.

An outbreak has swept through a city and the city has been sealed off and left to rot. A handful of survivors have fortified a base at the edge of a ruined district: a building, a few supply stations and a ring of light that the dead do not cross. Every night the dead return, and they do not come through the streets. They come up through the ground, from the metro tunnels and flooded basements beneath the city, so there is never a safe side to stand on.

Among the horde are stranger things: the Walker who was one of the first to turn, the Feral who moves like an animal, the Miner who was digging beneath the city when the outbreak began, and the Yaku, a tattooed gangster who died with a sword through his body and still has not stopped fighting.

The story is delivered through the environment and short text between waves rather than cutscenes, which keeps the game fast and the download small.

Endings & Progression

There is no final ending. The game is endless, and a run is measured by the highest wave reached. When every player is down, the run ends and the results card shows the stars, the best wave, the kills and the coins earned.

Progression carries between runs: the best wave and the star rating unlock new guns, which gives players a reason to come back for one more run.

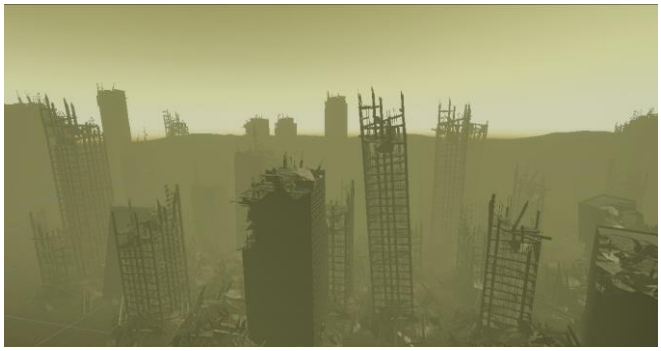


Figure 26: The ruined city: the district the players fight through.



Level Overview

The level is one continuous ruined city at night. The players start at the base building at the edge of the district and fight in a half-circle that opens into the city in front of it. The schematic below shows how the zones are laid out; it is not to scale.

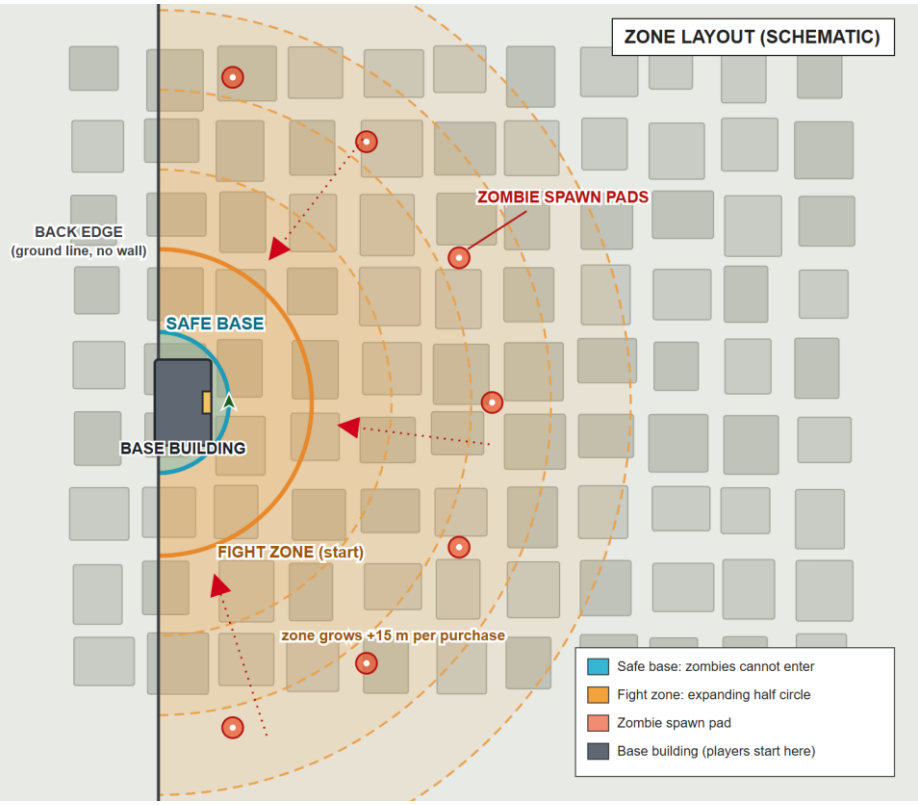


Figure 27: Zone layout (schematic).

Location Overviews

The Base

The base building stands against a straight back edge, a line on the ground at the rear of the zone, with no wall. In front of it is the half-circle safe base with a radius of 38 metres. The building carries the weapon racks and the armour, ammunition and medkit stations. A glowing ring on the ground marks the edge of the safe base so the players can always see where the safety ends.



The Fight Zone

In front of the safe base is the fight zone, a half circle that starts at a radius of 58 metres from the zone centre. The streets and ruined buildings inside it are the battlefield: buildings give cover and create choke points, open streets give room to run. Zombies rise out of the ground around the players wherever they fight inside it, so there is no safe corner of the zone.



Figure 28: In game: the cyan wall of the safe base around the base building, and the amber edge of the fight zone.

The Ruined City

Beyond the starting zone the district opens up into the rest of the city. The zone can grow to a radius of 298 metres, so the players gradually take more of the streets around the base as the run goes on.

Zone Expansion

Each expansion adds 15 metres to the radius of the fight zone and costs 20 coins. The area figures are those of the half circle.

Stage	Fight zone radius	Total coins spent	Playable area (half circle)
Start	58 m	0	5,332 m²
1	73 m	20	8,431 m²
2	88 m	40	12,236 m²
4	118 m	80	21,968 m²
8	178 m	160	49,915 m²
12	238 m	240	89,171 m²
16 (maximum)	298 m	320	139,737 m²

The safe base never grows. Only the fight zone does, so every expansion gives the players more room to fight and retreat in, while the zombies keep rising around them wherever they go.



Market Analysis

Target Market

Zombie Rush targets two audiences with one game. On PC, it is aimed at players who enjoy short, replayable co-op shooters and horde modes. On Android, it is aimed at the large audience for zombie shooters on phones, who want a game that can be played in a few minutes, with controls that work on a touch screen.

The core age range is 13 to 30. The violence is fantasy horror against zombies rather than people, which suits a Teen-style rating on the store. The simple rules and short runs suit mobile play, while the rising difficulty, the zone economy and squad co-op give PC players a reason to master it.

Top Performers

The zombie wave-survival format has proven itself many times over. The Zombies mode of the Call of Duty series built a lasting audience on exactly this loop of surviving rounds, earning points from kills and spending them to open up the map. Killing Floor 2 showed how well a co-op wave shooter with an economy between waves works on PC. On mobile, Dead Trigger 2 and Into the Dead 2 showed that zombie shooting with simplified controls can reach a very large audience.

Zombie Rush takes the round-based structure and map-opening economy of these games and fits it into a format small and quick enough for a phone, without losing the tension that makes the genre work.

Feature Comparison

Feature	Zombie Rush	Call of Duty Zombies	Killing Floor 2
Wave-based survival	Yes	Yes	Yes
Currency earned from kills	Yes (coins)	Yes (points)	Yes (dosh)
Currency opens more of the map	Yes (zone radius)	Yes (doors and barriers)	No
Fixed safe base zombies cannot enter	Yes	No	No
Weapon tiers and progression	Planned	Yes	Yes
Online co-op	Planned	Yes	Yes
Built for phones as well as PC	Yes	No	No



The main differences are the safe base and the expanding fight zone. Instead of opening doors one by one, Zombie Rush grows the whole playable area outward in rings from a base the zombies cannot enter, which is easy to understand at a glance and works on a small screen.



Figure 29: Gameplay as wave 1 begins, with the full HUD and the rifle in hand.

Monetization

Zombie Rush is free to play and earns money in two ways: in-game purchases and ads. The **in-game purchases** are the special weapons, the Machine Gun and the Shotgun, which players can buy in the in-game store. The progression guns and the grenade can only be earned in play, and are never sold. The **ads** are small banner ads in the menus and an optional **revive ad**: when a player dies, they can watch a short ad to come back to life and keep playing.



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